

SUMMARY

Full Stack Developer with experience in designing scalable backend systems, building APIs, and optimizing data-driven applications. Proficient in Java, Python, SQL, and modern web technologies like React. Experienced in developing and integrating RESTful services, improving system performance, and working in agile environments. Skilled in integrating AI-powered features into modern software systems using OpenAI APIs and intelligent automation workflows.

TECHNICAL SKILLS

- Languages: Java, Python, JavaScript/TypeScript, SQL
- Frontend & Backend: React.js, Node.js, Spring Boot, REST APIs, Microservices
- Databases: MySQL, MongoDB
- Cloud & DevOps: AWS, Docker, Kubernetes (Basics), CI/CD
- Tools: Git, Postman, Kafka
- Concepts: Data Structures, OOP, System Design
- AI Integration: OpenAI APIs, Prompt Engineering

WORK EXPERIENCE

Tata Consultancy Services, System Engineer, Noida, India

Jul, 2023 – Present

- Designed and developed scalable backend services, REST APIs, and microservice-based modules supporting enterprise applications handling large-scale data.
- Integrated and managed 100+ REST APIs to enable seamless communication between frontend and backend systems.
- Optimized SQL queries and database performance, improving response time by 35%.
- Built and maintained backend modules and real-time data pipelines for monitoring, analytics, and production support systems.
- Collaborated with frontend teams to ensure smooth API integration and user experience.
- Worked in Agile environment, participating in sprint planning, code reviews, and deployments.
- Integrated AI-powered features using OpenAI APIs for automated content generation and intelligent data processing.
- Leveraged prompt engineering techniques to enhance system capabilities and improve user interaction workflows.

PROJECTS & INTERNSHIP

Applied Machine Learning and Data Science for Real-World Optimization Problem ([Research Chapter link](#))

Sep, 2024

- Engineered **supervised learning and time-series forecasting models** (Random Forest, Gradient Boosting, and LSTM) to predict crop yield, soil moisture, and irrigation demand with **>90% accuracy**.
- Developed a **real-time data acquisition pipeline** integrating IoT sensors and cloud databases for continuous data ingestion and model retraining.
- Applied **multivariate regression and feature selection** to identify key environmental parameters influencing productivity.
- Proposed a **reinforcement learning-based optimization framework** that dynamically adjusted irrigation schedules, reducing water consumption by **25%** and pesticide usage by **40%**.
- The study demonstrated a scalable approach to **data-driven decision-making** in resource-constrained environments applicable to other industrial and sustainability domains.

AICTE NEAT, Data Analyst, Moradabad, India

Dec, 2022 – Jan, 2023

- Processed and analyzed large-scale dairy production data from multiple facilities to identify inefficiencies and optimization opportunities.
- Implemented **data preprocessing, outlier detection, and feature extraction** techniques to ensure **95% data accuracy** across multiple datasets.
- Automated complex data workflows using **Python (Pandas, NumPy)** and **Selenium-based scripts**, reducing data processing time from **45 days to 7 days**.
- Applied **regression modeling and trend analysis** to uncover actionable insights for resource allocation and process improvement.
- Produced **interactive analytical reports and visual dashboards**, aiding data-driven decision-making and strategic planning.

Intelligent Biometric Authentication System using Deep Learning

Nov, 2023

- Designed and implemented a **Convolutional Neural Network (CNN)** for fingerprint classification and feature extraction, achieving **98% verification accuracy** across 100,000+ fingerprint samples.
- Introduced a **Siamese Network architecture** to perform **pairwise fingerprint similarity scoring**, enabling robust real-time identity verification and preventing duplicate entries.
- Enhanced model performance using **data augmentation** and **batch normalization**, reducing overfitting and improving generalization on unseen data.
- Integrated the trained DL model into a secure **Flask-based API** for real-time authentication, combining backend optimization and ML deployment pipelines.

EDUCATION

Dr. APJ Abdul Kalam Technical University, Lucknow, India	Jul, 2023
B.Tech. in Information Technology (8.4/10)	
Bonne Anne Public School, Moradabad, India	Apr, 2019
12th Science (75.4%)	

LEADERSHIP EXPERIENCE

Technical Club, President, Greater Noida, India	Jul, 2021 – Mar, 2023
- Led a team of 20+ members and organized 5+ technical events benefiting 300+ students. - Directed hackathons and technical activities, securing sponsorships and industry engagement opportunities.	

CERTIFICATIONS

Career Essentials in Data Analysis, Microsoft, and LinkedIn	May, 2025 – July, 2025
Microsoft Security Essentials, Microsoft, and LinkedIn	Jun, 2025 – July, 2025
The Basics of Google Cloud Compute, Google Cloud	Aug, 2024 – Sep, 2024
Supervised Machine Learning: Regression and Classification, Coursera	Oct, 2022 – Dec, 2022